The Flutter of a Uniform Cantilever Wing

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The flutter speed of a uniform cantilever wing (one with a built-in root and with constant mass and stiffness characteristics along its entire span) is determined by integration of the differential equations for the wing motion. The solution is free of the dynamical approximations usually made during flutter analyses of the Rayleigh-Ritz type. A brief discussion of the procedures conventionally employed for the solution of the flutter problem is given, followed by a detailed derivation of the exact method for finding the flutter speed. It is found that a Rayleigh-Ritz type analysis, based on the fundamental uncoupled bending and torsion modes, is entirely adequate for the flutter analysis of a uniform wing. In addition to yielding a flutter speed which is in excellent agreement with the true value, the approximate method adequately predicts the wing shape at flutter. Additional parts of the paper deal with the relation between the uncoupled and the ground modes for the uniform wing, and with the nature of the wing response resulting from forced excitation while in flight.

Introduction

A necessary step in the design of modern aircraft is an investigation of the flutter characteristics of the intended structure. Flutter, an aeroelastic instability, must not occur within the operating speed range of the craft, as it may cause catastrophic disintegration of the airplane during flight and will, in any event, involve serious oscillatory distortions of the structural components. Present design trends are in the direction of structures particularly vulnerable to flutter-type instabilities, thus intensifying the interest of aircraft designers in accurate methods for flutter analysis.

Briefly, the condition commonly referred to as flutter is a self-sustained oscillation which involves a coupling between the inertia, elastic, and damping forces of the vibrating airplane structure, together with the air forces acting on the aerodynamic surfaces. For flight at low speeds it is found that free oscillations of the airframe are damped away through the action of the air forces and structural damping. At the flutter speed a condition of neutral stability is attained, while at slightly higher air speeds the oscillations are of divergent character. It should be noted that other dynamic instabilities besides those of the aeroelastic type can occur in aircraft systems; for example, rigid-body motions of the craft can couple with the aerodynamic forces, causing a condition of "dynamic instability" to exist. This differs from flutter in that it involves motion of the craft as a rigid body, save for the deflections of control surfaces; relative distortions of the structure, an essential part of flutter, are not taken into account.

It is readily recognized that the solution of the flutter problem involves consideration of the total airplane structure as a continuous vibrating system acted on by the external air forces and internal damping. Because of the complexity of modern aircraft, the designer is forced to discard this broad point of view in favor of a simpler one. Accordingly, flutter is confined to either "wing flutter" or "tail flutter," assuming that the tail does not participate to an important extent in free wing vibrations and vice versa. For wing flutter, only wing and fuselage motions are taken into account. The fuselage may be assumed to act as a rigid mass, or fuselage distortions may be considered. Accurate tail-flutter analysis almost always requires that the effects of fuselage torsion and bending be accounted for.

As a second simplifying assumption, it is usual to describe the participation of the various airplane components in the flutter motion in terms of a few arbitrarily chosen "modes." In this manner the continuous airplane structure is replaced by a simpler system composed of a finite number of degrees of freedom. The arbitrary modes are usually taken as the fundamental modes of portions of the structure vibrating as an undamped system in vacuo. Thus the first few normal modes of a wing-fuselage structure are sometimes used to describe the flutter motion of an airplane. An even greater simplification of the flutter analysis is effected by using as degrees of freedom the fundamental "uncoupled" modes for the wing vibrating in vacuo; an uncoupled bending mode is arrived at by applying frictionless constraints to the wing which allow it to bend vertically but not twist about its elastic axis; for an uncoupled-torsion mode, frictionless constraints are applied which suppress the wing bending but allow it to twist. Analogous assumptions are made for tail-flutter analysis, in order to reduce the work required to a reasonable amount.

Since exact solutions for the flutter speeds of aircraft are not available, the extent of the error introduced by these simplifying assumptions is not known. Wind-tunnel flutter tests and other types of test have not thus far yielded accurate comparisons of the calculated and true flutter speeds. Considerable insight into the question is obtained from the discussions of classical dynamics, but the conclusions of classical dynamics, dealing with vibrations in vacuo, are largely inapplicable to the flutter problem (see later discussion).

The present paper is a study of the validity of approximating the flutter motion of a continuous wing by combinations of a few of its uncoupled modes for vibration in vacuo. To do this, an exact solution is obtained for the flutter speed of a rectangular wing with a uniform mass and stiffness distribution along its span; mathematical difficulties prevent the choice of a wing of more complex design. The true flutter speed for the uniform wing is then compared with the speeds calculated by the more approximate methods previously discussed.

In succeeding sections of the paper an outline of the simplified methods for flutter analysis is given, together with the exact solution for the flutter of a uniform cantilever wing. Additional notes are concerned with the problems of forced oscillations in flight and the relation between the normal modes and the uncoupled modes for the uniform wing.

Flutter Analysis by Conventional Methods

Several authors (1, 2, 3)[2] have discussed the problem of analyzing conveniently the flutter characteristics of airplane struc-

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NOTE: Statements and opinions advanced in papers are to be understood as individual expressions of their authors and not those of the Society.

[2] Numbers in parentheses refer to the Bibliography at the end of the paper.

tures. The methods now in most common use incorporate the simplifications mentioned in the "Introduction" and are carried out according to a modified Rayleigh-Ritz procedure. The calculation of the flutter speed of a uniform cantilever wing will now be outlined, enabling the reader to become acquainted with the various aspects of the solution.

It will be assumed that the flutter motion of the wing can be described by a combination of the fundamental uncoupled modes in bending and torsion for the wing vibrating in vacuo. Since the wing has a uniform distribution of mass and stiffness along its span, these become the fundamental bending and torsion modes of a uniform bar, whose properties are well known (4). The natural frequency of the fundamental bending mode is

$$\omega_b = \frac{3.55}{l^2}\sqrt{\frac{EI_b}{m}} \quad [1]$$

and the corresponding mode shape is

$$Y_b = C_b\{\cosh \beta x - 0.734 \sinh \beta x - \cos \beta x + 0.734 \sin \beta x\} \quad [2]$$

where

EI_b = flexural stiffness of wing
m = wing mass per unit length of span
l = cantilever wing length
βl = 0.6π

and C_b is an arbitrary constant and x is a spanwise co-ordinate with its origin at the wing root. For the fundamental torsion mode, the natural frequency is

$$\omega_t = \frac{\pi}{2l}\sqrt{\frac{GJ}{I}} \quad [3]$$

and the mode shape is

$$Y_t = C_t \sin \frac{\pi}{2}\frac{x}{l} \quad [4]$$

where

GJ = torsional stiffness of wing
I = mass moment of inertia of wing per unit length of span
C_t = arbitrary constant

The continuous wing structure is thus replaced by a system composed of two degrees of freedom. The wing-bending shape is taken to be

$$y = y_0(t)\{\cosh \beta x - 0.734 \sinh \beta x - \cos \beta x + 0.734 \sin \beta x\} \quad [5]$$

and the torsion is approximated as

$$\theta = \theta_0(t)\left\{\sin \frac{\pi}{2}\frac{x}{l}\right\} \quad [6]$$

where $y_0(t)$ and $\theta_0(t)$ are the generalized co-ordinates, both functions only of the time t, which make up the two degrees of freedom of the system. The equations of motion for the wing in terms of $y_0(t)$ and $\theta_0(t)$ can now be deduced without difficulty by using Lagrange's Equations [5].

The Lagrange equations for the problem are

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{y}_0} + \frac{\partial V_E}{\partial y_0} = L_G \quad [7]$$

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{\theta}_0} + \frac{\partial V_E}{\partial \theta_0} = M_G \quad [8]$$

where the use of a dot refers to a time derivative, and

T = kinetic energy of system
V_E = potential energy of system
L_G = generalized aerodynamic lift force corresponding to bending mode
M_G = generalized aerodynamic moment corresponding to torsion mode

(These equations do not account for the presence of structural damping. If this effect is to be considered, additional terms involving a dissipation function must be introduced in Equations [7] and [8]. None of the conclusions of this paper is dependent to any extent on the presence or absence of damping; accordingly, for the sake of simplicity, the structural damping terms are omitted.)

In terms of the wing bending y and the torsion θ, both referred to the elastic axis, the kinetic energy of the system is

$$2T = m\int_0^l \dot{y}^2 dx + 2S\int_0^l \dot{y}\dot{\theta}dx + I\int_0^l \dot{\theta}^2\, dx \quad [9]$$

and the potential energy is

$$2V_E = EI_b\int_0^l\left(\frac{\partial^2 y}{\partial x^2}\right)^2 dx + GJ\int_0^l\left(\frac{\partial \theta}{\partial x}\right)^2 dx \quad [10]$$

where x is a spanwise co-ordinate with its origin at the wing root, and S is the static mass unbalance of the wing about its elastic axis, per unit of span length. The static unbalance is taken as positive for a chordwise center-of-gravity location which is behind the elastic axis.

Substituting Equations [5] and [6] in Equations [9] and [10] and performing the necessary integrations, the expressions for T and V_E become

$$2T = 1.016\, ml\dot{y}_0^2 + 1.368\, Sl\dot{y}_0\dot{\theta}_0 + 0.500\, Il\dot{\theta}_0^2 \quad [11]$$

and

$$2V_E = 12.800\, EI_b\frac{y_0^2}{l^3} + 1.232\, GJ\frac{\theta_0^2}{l} \quad [12]$$

The forms of the generalized lift force L_G and the generalized moment M_G are arrived at as follows: Let (δy_0) be an arbitrary displacement of the co-ordinate $y_0(t)$. The work done during this displacement is $L_G(\delta y_0)$. Corresponding to the displacement δy_0, let the wing deflection at a given station be δy. Then if Ldx is the aerodynamic lift force acting at that station over a small length of span dx, the work done on the increment of span is $(Ldx)\delta y$. Equating the works written in terms of the generalized and actual forces and displacements, the necessary relation is derived

$$L_G(\delta y_0) = \int_0^l L(\delta y)dx$$

or

$$L_G = \int_0^l L\left(\frac{\partial y}{\partial y_0}\right)dx \quad [13]$$

Here L is the lift force acting per unit of span length at station x. In a similar manner the expression for M_G is found to be

$$M_G = \int_0^l M\left(\frac{\partial \theta}{\partial \theta_0}\right)dx \quad [14]$$

where M is the aerodynamic moment acting about the elastic axis at station x.

If consideration is restricted to the condition when the wing moves through the air at the flutter speed so that a disturbance causes the wing to oscillate sinusoidally and with constant amplitude, the air forces derived by Theodorsen (6) can be used to describe L and M. His work shows that L and M may be taken in the form

$$L = \omega^2 L_y y + \omega L'_y \dot{y} + \omega^2 L_\theta \theta + \omega L'_\theta \dot{\theta} \quad [15]$$

$$M = \omega^2 M_y y + \omega M'_y \dot{y} + \omega^2 M_\theta \theta + \omega M'_\theta \dot{\theta} \quad [16]$$

where ω is the circular frequency of the sinusoidal wing oscillation, and the coefficients L_y, L'_y, ..., M_θ, M'_θ are functions of the wing semichord length b, the elastic-axis location, the density of the air at rest, the circular frequency ω, and the true air speed V. In particular, the coefficients are functions of the "reduced frequency" $\omega b/V$, and for a particular wing and air density, their values remain constant so long as the reduced frequency does not change.

It is readily shown from Theodorsen's work that the coefficients L_θ and M_y are not equal, and that L'_θ does not equal M'_y. This implies that the air forces acting on the oscillating wing are not derivable from a potential function. The fact that the air forces cannot be treated in a simple manner, in terms of a potential function, considerably adds to the complication of the general flutter problem.

It is to be noted further that Equations [15] and [16] are statements of the fundamental concept that the lift force and moment acting on an oscillating airfoil are not in phase with the displacements. The fact that the coefficients of the type L_y are only functions of the reduced frequency (subject to the conditions mentioned) is important to the later solution.

Equations [5] and [6] can now be substituted in Equations [15] and [16] (remembering that the oscillations are sinusoidal), and the resulting expressions can then be put in Equations [13] and [14]. If this is done, the forms for the generalized force L_G and moment M_G are found to be, after appropriate integration

$$L_G = 1.016\, l\omega^2 L_y y_0 + 1.016\, l\omega L'_y \dot{y}_0 + 0.684\, l\omega^2 L_\theta \theta_0 + 0.684\, l\omega L'_\theta \dot{\theta}_0 \quad [17]$$

and

$$M_G = 0.684\, l\omega^2 M_y y_0 + 0.684\, l\omega M'_y \dot{y}_0 + 0.500\, l\omega^2 M_\theta \theta_0 + 0.500\, l\omega M'_\theta \dot{\theta}_0 \quad [18]$$

Putting the expressions for T, V_E, L_G, and M_G in the Lagrange Equations [7] and [8], the equations of motion for the wing are obtained in terms of y_0 and θ_0. They are, after slight simplification

$$m\ddot{y}_0 - \omega L'_y \dot{y}_0 - \omega^2 L_y y_0 + 12.6 \frac{EI_b}{l^4} y_0 + 0.673\, S\ddot{\theta}_0 - 0.673\, \omega L'_\theta \dot{\theta}_0 - 0.673\, \omega^2 L_\theta \theta_0 = 0 \quad [19]$$

$$S\ddot{y}_0 - \omega M'_y \dot{y}_0 - \omega^2 M_y y_0 + 0.772\, I\ddot{\theta}_0 - 0.772\, \omega M'_\theta \dot{\theta}_0 - 0.772\, \omega^2 M_\theta \theta_0 + 1.900 \frac{GJ}{l^2} \theta_0 = 0 \quad [20]$$

It should be recalled that these equations apply only at the flutter condition; the air forces have not been introduced in sufficiently general form to account for possible convergent or divergent oscillations.

The equations of motion are solved in order to determine the flutter speed for the wing according to the following line of reasoning: Suppose that y_0 and θ_0 are solutions to Equations [19] and [20]. Now, multiply Equations [19] and [20] by the imaginary quantity i and let $\bar{y}_0$ and $\bar{\theta}_0$ be solutions of the resulting pair of equations. If Equation [19] is added to the first of the two newly formed equations, and Equation [20] is added to the second, two new equations result

$$m\ddot{Y}_0 - \omega L'_y \dot{Y}_0 - \omega^2 L_y Y_0 + 12.6 \frac{EI_b}{l^4} Y_0 + 0.673\, S\ddot{\Theta}_0 - 0.673\, \omega L'_\theta \dot{\Theta}_0 - 0.673\, \omega^2 L_\theta \Theta_0 = 0 \quad [21]$$

$$S\ddot{Y}_0 - \omega M'_y \dot{Y}_0 - \omega^2 M_y Y_0 + 0.772\, I\ddot{\Theta}_0 - 0.772\, \omega M'_\theta \dot{\Theta}_0 - 0.772\, \omega^2 M_\theta \Theta_0 + 1.900 \frac{GJ}{l^2} \Theta_0 = 0 \quad [22]$$

where

$$Y_0 = y_0 + i\bar{y}_0 \quad [23]$$

$$\Theta_0 = \theta_0 + i\bar{\theta}_0 \quad [24]$$

Inspection of Equations [21] and [22] shows that Y_0 and Θ_0 can be taken of the form

$$Y_0 = \bar{A}e^{i\omega t} \quad [25]$$

$$\Theta_0 = \bar{B}e^{i\omega t} \quad [26]$$

where $\bar{A}$ and $\bar{B}$ are complex constants. It is clear from the foregoing discussion that y_0 is the real part of expression [25], while θ_0 is the real part of the expression [26].

Substituting Equations [25] and [26] for Y_0 and Θ_0 in Equations [21] and [22] gives, after slight manipulation

$$(L_{11} - iL'_y)\bar{A} - 0.673(L_{12} + iL'_\theta)\bar{B} = 0 \quad [27]$$

$$-(M_{21} + iM'_y)\bar{A} + 0.772(M_{22} - iM'_\theta)\bar{B} = 0 \quad [28]$$

where

$$L_{11} = m\left(\frac{\omega_b^2}{\omega^2} - 1\right) - L_y; \quad L_{12} = S + L_\theta$$

$$M_{21} = S + M_y; \quad M_{22} = I\left(\frac{\omega_t^2}{\omega^2} - 1\right) - M_\theta$$

From this it is seen that if nontrivial solutions for Y_0 and Θ_0 are to exist, the determinant formed by the coefficients of $\bar{A}$ and $\bar{B}$ in Equations [27] and [28] must vanish. This determinant has complex elements and in expanded form, the statement of its vanishing is

$$(\Delta_1 - \Delta_2) + i(\Delta_3 + \Delta_4) = 0 \quad [29]$$

where

$$\Delta_1 = \begin{vmatrix} L_{11} & -0.673L_{12} \\ -M_{21} & 0.772M_{22} \end{vmatrix}; \quad \Delta_2 = \begin{vmatrix} -L'_y & -0.673L'_\theta \\ -M'_y & -0.772M'_\theta \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} -L'_y & -0.673L_{12} \\ -M'_y & 0.772M_{22} \end{vmatrix}; \quad \Delta_4 = \begin{vmatrix} L_{11} & -0.673L'_\theta \\ -M_{21} & -0.772M'_\theta \end{vmatrix}$$

It is clear, then, if a flutter speed for the wing does exist, Equation [29] will be satisfied for a particular combination of ω and $\omega b/V$ values. Physical conditions require, of course, that both the circular and reduced frequencies have real values at the flutter point.

Accordingly, the final step in the solution for the flutter speed may be stated as follows: To find a combination of the circular frequency ω and the reduced frequency $\omega b/V$, both real quantities, such that the stability Equation [29] is satisfied.

The mechanics for carrying out this last step will not be discussed here, and reference is made to the work of several other authors (6, 8, 9). One obvious method used by Theodorsen (6) is to investigate the combination of frequencies which cause the real and imaginary parts of Equation [29] to vanish simultaneously.

It is evident that once the proper combination of circular and reduced frequencies for flutter are determined, the value of the flutter speed can be readily determined, thus completing the solution.

In the solution of physical problems it is clear that the introduction of complex functions is not a necessity, although it is frequently a mathematical convenience. Some interest is attached to the manner in which the solution to Equations [19] and [20] proceeds without the introduction of complex forms. A brief discussion of this alternate method of solution is given in the Appendix.

Exact Solution for Flutter of Uniform Wing

To arrive at the exact solution for the flutter of a uniform wing, which is to be compared with the results of the approximate method described in the last section, it is first necessary to deduce the differential equations which govern the motion of the wing.

At station x the wing has a bending displacement $+y$ downward and a twisting displacement $+\theta$ nose-up, both referred to the chordwise position of the elastic axis (see Fig. 1). Consider now, the dynamic equilibrium of a small increment of wing length dx located in the vicinity of this station. The inertia force $(m\ddot{y} + S\ddot{\theta})dx$ and the inertia moment $(I\ddot{\theta} + S\ddot{y})dx$ must be supplied by the external forces and moments, as shown in Fig. 2.[3] The external forces and moments consist of transverse shearing forces and torsional moments which are transmitted from one portion of the wing to the next, plus the aerodynamic lift force and pitching moment.

It is shown in the standard texts on strength of materials (10) that the transverse shearing force N acting upward at station x, on the inboard side of the increment of wing length, is

$$N = -EI_b \frac{\partial^3 y}{\partial x^3} \qquad [30]$$

where the partial derivative is used since the deflection y is a function of both x and the time t. On the outboard side of the wing element, at station $x + dx$, the downward transverse shear is then equal to $N + \dfrac{\partial N}{\partial x}\, dx$.

The nose-down torsional moment acting on the inboard side of the wing element is given by (11)

$$T = GJ \frac{\partial \theta}{\partial x} \qquad [31]$$

At station $x + dx$ the torsional moment is $T + \dfrac{\partial T}{\partial x}\, dx$. Again, partial derivatives are indicated since θ is a function of both x and t.

The nature of the aerodynamic forces acting on the increment of wing length at the flutter condition has already been discussed in the preceding section of the paper. Equations [15] and [16] describe the lift force per unit span, positive when downward, and the pitching moment per unit span, acting nose-up about the elastic axis, in terms of the deflections y and θ and their associated velocities. Over a wing length dx the lift force is then Ldx and the pitching moment equals Mdx.

According to Fig. 2, the equations of motion for the wing become

$$m\ddot{y} + S\ddot{\theta} + EI_b \frac{\partial^4 y}{\partial x^4} - \omega^2 L_y y - \omega L'_y \dot{y} - \omega^2 L_\theta \theta - \omega L'_\theta \dot{\theta} = 0 \qquad [32]$$

and

$$I\ddot{\theta} + S\ddot{y} - GJ \frac{\partial^2 \theta}{\partial x^2} - \omega^2 M_y y - \omega M'_y \dot{y} - \omega^2 M_\theta \theta - \omega M'_\theta \dot{\theta} = 0 \qquad [33]$$

where, it is recalled, the dots represent time derivatives and for a specific wing and air density, the coefficients L_y, L'_y, ..., M_θ, M'_θ are functions only of the reduced frequency $\omega b/V$. Since the air forces of Equations [15] and [16] are for a wing oscillating sinusoidally with a constant amplitude, it should be noted that Equations [32] and [33] apply only at the flutter condition.

The equations of motion, Equations [32] and [33], can be solved by taking

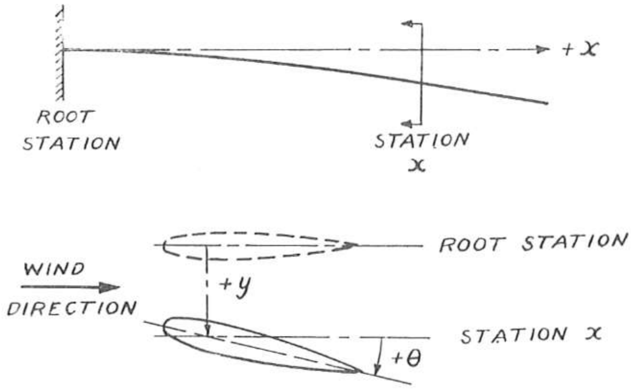


Fig. 1 Co-Ordinates Used in Discussion of Uniform Wing

$$y = Y_1(x) \cos(\omega t + \psi) + Y_2(x) \sin(\omega t + \psi) \qquad [34]$$

$$\theta = Y_3(x) \cos(\omega t + \psi) + Y_4(x) \sin(\omega t + \psi) \qquad [35]$$

where ψ is an arbitrary phase angle, and $Y_1(x)$, $Y_2(x)$, $Y_3(x)$ and $Y_4(x)$ are functions of the span co-ordinate x only and describe the wing shape at flutter. The forms of the four Y functions are yet to be determined.

Substituting Equations [34] and [35] in the equations of motion, separating the terms in each equation which have $\sin(\omega t + \psi)$ and $\cos(\omega t + \psi)$ affixed, and equating each set individually to zero, the four equations are obtained

$$EI_b Y_1^{\mathrm{IV}} - m\omega^2 Y_1 - \omega^2 L_y Y_1 - \omega^2 L'_y Y_2 - S\omega^2 Y_3 - \omega^2 L_\theta Y_3 - \omega^2 L'_\theta Y_4 = 0 \qquad [36]$$

$$\omega^2 L'_y Y_1 + EI_b Y_2^{\mathrm{IV}} - m\omega^2 Y_2 - \omega^2 L_y Y_2 + \omega^2 L'_\theta Y_3 - S\omega^2 Y_4 - \omega^2 L_\theta Y_4 = 0. \qquad [37]$$

$$\omega^2 S Y_1 + \omega^2 M_y Y_1 + \omega^2 M'_y Y_2 + GJ Y_3^{\mathrm{II}} + I\omega^2 Y_3 + \omega^2 M_\theta Y_3 + \omega^2 M'_\theta Y_4 = 0 \qquad [38]$$

$$\omega^2 M'_y Y_1 - \omega^2 S Y_2 - \omega^2 M_y Y_2 + \omega^2 M'_\theta Y_3 - GJ Y_4^{\mathrm{II}} - I\omega^2 Y_4 - \omega^2 M_\theta Y_4 = 0 \qquad [39]$$

where the Roman numeral superscripts denote differentiations with respect to x.

These four equations serve to fix the forms of the four Y functions. They are taken to be linear combinations of sets of terms of the type

$$Y_1 = Ae^{\lambda x} \qquad [40]$$

$$Y_2 = Be^{\lambda x} \qquad [41]$$

$$Y_3 = Ce^{\lambda x} \qquad [42]$$

$$Y_4 = De^{\lambda x} \qquad [43]$$

If this typical set of Y forms is substituted in Equations [36] to [39], the four equations result

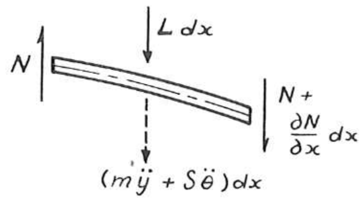
$$\left[\lambda^4 \frac{EI_b}{\omega^2} - (m + L_y)\right] A - L'_y B - (S + L_\theta) C - L'_\theta D = 0. \qquad [44]$$

$$L'_y A + \left[\lambda^4 \frac{EI_b}{\omega^2} - (m + L_y)\right] B + L'_\theta C - (S + L_\theta) D = 0. \qquad [45]$$

$$(S + M_\theta) A + M'_\theta B + \left[\lambda^2 \frac{GJ}{\omega^2} + (I + M_\theta)\right] C + M'_\theta D = 0 \qquad [46]$$

$$-M'_\theta A + (S + M_\theta) B - M'_\theta C + \left[\lambda^2 \frac{GJ}{\omega^2} + (I + M_\theta)\right] D = 0 \qquad [47]$$

[3] Notation introduced in the last section is carried over without change to this discussion.



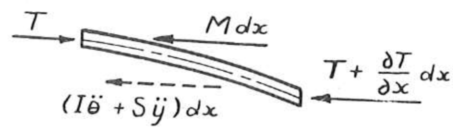


FIG. 2(a) FORCES ACTING ON AN ELEMENT OF WING OF LENGTH dx FIG. 2(b) MOMENTS ACTING ON AN ELEMENT OF WING OF LENGTH dx
(The arrows represent moment vectors in the usual right-hand sense.)

It is then clear that if the chosen Y forms are valid, the set of four homogeneous Equations [44] to [47] must be satisfied. In other words, the determinant formed by the coefficients attached to A, B, C, D in these equations must equal zero. The meaning of this requirement, in physical terms, will be given later.

Consider now the boundary conditions for the problem. Since the root of the uniform wing is built in, it must be true that for all values of the time t

$$\text{at } x = 0, \qquad y = 0 \qquad [48]$$

$$\frac{\partial y}{\partial x} = 0 \qquad [49]$$

$$\theta = 0 \qquad [50]$$

At the wing tip there can be no bending moment, transverse shearing force, or torsional moment acting. Therefore for all values of t, it is necessary that

$$\text{at } x = l, \qquad \frac{\partial^2 y}{\partial x^2} = 0 \qquad [51]$$

$$\frac{\partial^3 y}{\partial x^3} = 0 \qquad [52]$$

$$\frac{\partial \theta}{\partial x} = 0 \qquad [53]$$

where l is the span of the cantilever wing. In addition to these boundary conditions, it must also be possible to adjust the solution so that at the start of the motion, when $t = 0$, an arbitrary deflection and velocity can be assigned to some point of the wing.

Inspection of Equations [34] and [35] shows that the boundary conditions [48] to [53] are equivalent to the requirement that

$$\text{at } x = 0, \quad Y_1 = Y_1^{\text{I}} = Y_2 = Y_2^{\text{I}} = Y_3 = Y_4 = 0 \qquad [54]$$

and

$$\text{at } x = l, \quad Y_1^{\text{II}} = Y_1^{\text{III}} = Y_2^{\text{II}} = Y_2^{\text{III}} = Y_3^{\text{I}} = Y_4^{\text{I}} = 0 \qquad [55]$$

These conditions must be satisfied by the Y functions when they are written in complete form, as is done later in Equations [56] to [59].

It will now be shown that the determinantal equation which arises from Equations [44] to [47], combined with the conditions [54] and [55], completely fix the flutter speed and the flutter shape of the uniform cantilever wing. The shape at flutter will be known, but a single amplitude constant will be indeterminate. The value of this constant, together with that of the phase angle ψ, will be fixed once a starting deflection and velocity are assigned to a point of the wing.

Returning now to a consideration of the determinantal equation arising from the relations Equations [44] to [47], it is seen that this equation is of the sixth degree in λ^2. Moreover, since the flutter frequency ω is of necessity a real number, all terms other than λ in the coefficients of Equations [44] to [47] are real. Hence the coefficients of the sixth-degree equation are also real. There are then six λ^2 roots, separating into three pairs of complex conjugates. The corresponding λ values are twelve in number, dividing into six pairs of complex conjugates, i.e., three sets of conjugates plus their negatives.

(It is of interest to note that the determinant formed by the coefficients of Equations [44] to [47] can be written in a form similar to that of Equation [84]. Equation [85] then follows. One of the brackets of this last equation, when equated to zero, yields three distinct λ^2 values; the complex conjugates of these roots are obtained if the remaining bracket is set equal to zero.)

The complete expressions for the Y functions, as outlined in Equations [40] to [43], can now be written. Since there are twelve possible values for λ, the complete forms become

$$Y_1 = \sum_r \{A_r e^{\lambda_r x} + A_{r+1} e^{\lambda_{r+1} x}\} \qquad [56]$$

$$Y_2 = \sum_r \{B_r e^{\lambda_r x} + B_{r+1} e^{\lambda_{r+1} x}\} \qquad [57]$$

$$Y_3 = \sum_r \{C_r e^{\lambda_r x} + C_{r+1} e^{\lambda_{r+1} x}\} \qquad [58]$$

$$Y_4 = \sum_r \{D_r e^{\lambda_r x} + D_{r+1} e^{\lambda_{r+1} x}\} \qquad [59]$$

where $\qquad r = 1, 3, 5, 7, 9, 11$

and λ_r and λ_{r+1} are complex conjugates. The ratios B_r/A_r, C_r/A_r, and D_r/A_r are determinate, once the value of λ_r is known, from Equations [44] to [47]. It can be shown without difficulty that pairs of coefficients of the type A_r and A_{r+1} are complex conjugates; hence if $B_r/A_r = \alpha_r + i\beta_r$, then $B_{r+1}/A_{r+1} = \alpha_r - i\beta_r$, and similarly for the other ratios.

Let

$$\lambda_r = \xi_r + i\eta_r; \quad \lambda_{r+1} = \xi_r - i\eta_r \qquad [60]$$

Equations [56] to [59] can then be written as

$$Y_1 = \sum_r e^{\xi_r x}[a_r \cos \eta_r x + a_{r+1} \sin \eta_r x] \qquad [61]$$

$$Y_2 = \sum_r e^{\xi_r x}[b_r \cos \eta_r x + b_{r+1} \sin \eta_r x] \qquad [62]$$

$$Y_3 = \sum_r e^{\xi_r x}[c_r \cos \eta_r x + c_{r+1} \sin \eta_r x] \qquad [63]$$

$$Y_4 = \sum_r e^{\xi_r x}[d_r \cos \eta_r x + d_{r+1} \sin \eta_r x] \qquad [64]$$

where

$$a_r = A_r + A_{r+1}; \quad a_{r+1} = i(A_r - A_{r+1}) \qquad [65]$$

$$b_r = B_r + B_{r+1}; \quad b_{r+1} = i(B_r - B_{r+1}) \qquad [66]$$

etc.

From this it is seen that if

$$B_r/A_r = \alpha_r + i\beta_r$$

then

$$b_r = \alpha_r a_r + \beta_r a_{r+1}$$

$$b_{r+1} = -\beta_r a_r + \alpha_r a_{r+1} \qquad [67]$$

and similarly for the remaining ratios.

If Equations [61] to [64] are now substituted in the boundary conditions, Equations [54] and [55], a set of twelve homogeneous equations in the coefficients a, b, c, d are obtained. Using results of the type Equation [67], the b, c, d coefficients can be eliminated, leaving a set of twelve homogeneous equations in the twelve coefficients a_r and a_{r+1}. Let the determinant formed by the coefficients of this set be Δ. Then for the Y forms of Equations [61] to [64] to be valid, it must be true that

$$\Delta = 0 \qquad [68]$$

Inspection of Δ shows that it is dependent on the values of λ, calculated in a previous step in the solution, and on the wing length l.

The mechanics of the solution for the flutter of a uniform wing can now be outlined. It has been shown that for any chosen values of the reduced frequency $\omega b/V$ and the circular frequency ω, a set of λ values can be calculated, using the determinantal equation arising from Equations [44] to [47]. Moreover, if the λ set is calculated for those values of $\omega b/V$ and ω at which wing flutter occurs, then this set will satisfy Equation [68]. Any set of λ values, corresponding to values of $\omega b/V$ and ω different from those at flutter, will result in a Δ value which is different from zero.

The problem of the flutter of a uniform wing is thus reduced to the following: To find a reduced frequency $\omega b/V$ and a flutter frequency ω such that the corresponding values of λ will cause Equation [68] to be satisfied. If the values of $\omega b/V$ and ω at flutter are determined, the flutter speed is at once calculable. The wing shape at flutter is defined by Equations [61] to [64]. It is readily seen that all but one of the coefficients a, b, c, d in these expressions can be computed from the set of twelve homogeneous equations previously discussed. The significance of the remaining coefficient has already been mentioned.

In conclusion, it is to be noted that there will be several flutter speeds for the wing; only the lowest of these is of practical importance.

Calculations for a 400-Mph Wing

The complexity of the flutter solutions outlined in the previous sections makes it difficult to draw general comparisons between the conventional and exact methods. Although some progress can be made in this direction (see "Discussion and Results"), questions such as how great an error the conventional methods introduce in the calculated flutter speed cannot be answered. It appears desirable then to choose a specific wing, of typical properties, for which numerical results can be obtained.

For this purpose a uniform cantilever wing having a flutter speed near 400 mph is a reasonable choice. The 400-mph flutter speed is representative of modern aircraft in the moderately high-speed class. Although study of a series of wings having flutter speeds ranging from very low to very high values would be useful, the enormous amount of computational labor involved places such an investigation beyond the scope of the present work.

The wing chosen for study in this paper has the following characteristics:

Cantilever wing span = 20 ft
Wing chord = 6 ft
Wing weight = 4 psf
Radius of gyration of wing about center of gravity = 25 per cent chord
Spanwise elastic axis of wing at 33 per cent chord (from leading edge)
Center of gravity of wing at 43 per cent chord (from leading edge)
Fundamental uncoupled wing bending frequency = ω_b = 50 radians per sec
Fundamental uncoupled wing torsion frequency = ω_t = 87 radians per sec

Based on these assigned properties, it is readily shown that

b = 3 ft	S = 0.447 slug ft per ft
m = 0.746 slugs per ft	EI_b/m = 31.7 × 10^6 lb ft^3 per slug
l = 20 ft	
I = 1.943 slug ft^2 per ft	GJ/I = 1.23 × 10^6 lb ft per slug

The values of the other parameters entering the solutions, such as L_y, L'_y, ..., M_θ, M'_θ, are functions of the reduced frequency. They are calculable from the results of Theodorsen (6), once a value of $\omega b/V$ of interest is chosen.

If a flutter analysis is carried out for this wing, using the "conventional" method outlined in the third section, it is found that flutter occurs at a reduced speed of 2.80, i.e., Equation [29] is satisfied at this value of $V/(b\omega)$.[4] The corresponding flutter speed and flutter frequency are found to be

$$V_f = 385 \text{ mph}; \quad \omega_f = 67.4 \text{ radians per sec}$$

To arrive at these results, any one of several known methods (6, 8, 9) can be used to solve the stability Equation [29].

In order to carry out a numerical flutter analysis for the wing by the exact method, a schedule for the computations had to be arranged. A review of the exact method shows that the line of attack suggested by the discussion in the last paragraphs of the preceding section is most fruitful. This involves a trial-and-error procedure (as do all known methods for determining flutter points).

In essence, the procedure is as follows: An arbitrary (real) value of $V/(b\omega)$ is chosen, together with an arbitrary (real) value of ω. From the determinant arising from Equations [44] to [47] the corresponding twelve values of λ are calculated. Using these λ, the value of the determinant Δ is found. If the computed value of Δ does not equal zero, as required by Equation [68], the procedure is repeated for a new set of $V/(b\omega)$ and ω values.

The amount of work required to determine the exact flutter point for a wing is considerably lessened by a judicious choice of trial values for the reduced speed and frequency. For the wing in question, it is natural to confine interest to the region surrounding the flutter point as computed by the approximate method, i.e., to the region near $V/(b\omega) = 2.80$ and $\omega = 67.4$ radians per sec.

The results of calculations for seven sets of values of $V/(b\omega)$ and ω are shown in Fig. 3. From this figure the flutter point can be located with sufficient accuracy for present purposes. Additional calculations were not carried out owing to the large amount of labor required for each Δ determination. It is to be recalled that in addition to considerable preliminary setup work, each Δ evaluation involves (a) finding the roots λ of a complex cubic (or a sixth-degree equation with real coefficients), (b) calculating the ratios of the type B_r/A_r, C_r/A_r, and D_r/A_r, and (c) determining the value of a 12 × 12 determinant with real elements (which arises from Y-function boundary conditions).

An inspection of Fig. 3 shows that flutter undoubtedly occurs quite close to the point[5]

[4] The reduced speed is the inverse of the reduced frequency.

[5] It is to be noted that slight error in interpolating the data of Fig. 3 to locate the flutter point does not appreciably affect the accuracy of the flutter-speed determination. Consider the extreme points $V/(b\omega) = 2.94$; $\omega = 65.7$ radians per sec, and $V/(b\omega) = 2.86$; $\omega = 66.7$ radians per sec. These points lie to either side of the flutter point described in the text. The values of V corresponding to these extreme points are 395 mph and 390 mph, respectively. These do not differ appreciably from the value 393 mph quoted for the wing.

TABLE 1 VALUES OF λ AND CONSTANTS a, b, c, d AT FLUTTER POINT FOR UNIFORM WING

r	λ_r	a_r	a_{r+1}	b_r	b_{r+1}	c_r	c_{r+1}	d_r	d_{r+1}
1	0.11184 — i0.00621	0.041	—0.097	—0.0971	—0.0407	—0.000661	0.005597	0.006121	0.000758
3	—0.11184 + i0.00621	0.386	—0.760	—0.7607	—0.3838	—0.008509	0.044819	0.048994	0.009587
5	0.01868 + i0.09976	0.008	0.356	0.3553	—0.0079	0.006773	0.060341	0.060287	—0.006694
7	—0.01868 — i0.09976	0.576	1.076	1.0741	—0.5745	0.114192	0.173986	0.173947	—0.113867
9	0.00530 + i0.08200	0.256	0.427	0.4234	—0.2504	—0.018893	0.130583	0.131049	0.020114
11	—0.00530 — i0.08200	—1.267	—1.000	—0.9955	+1.2459	—0.092881	—0.417561	—0.420376	0.089715

NOTE: The value of a_{12} has been arbitrarily set equal to —1.0000.

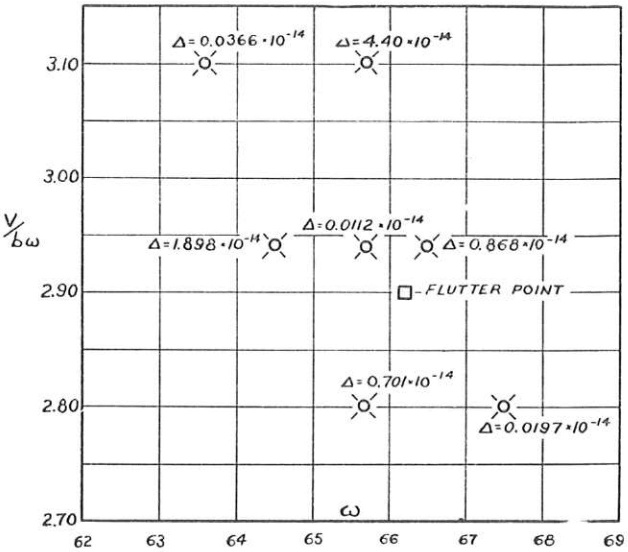


FIG. 3 VALUES OF THE DETERMINANT Δ FOR VARIOUS SETS OF VALUES $V/b\omega$ AND ω

(The flutter point has been located by interpolation of the Δ values.)

$$V/(b\omega) = 2.90; \quad \omega_f = 66.2 \text{ radians per sec}$$

which gives a flutter speed of

$$V_f = 393 \text{ mph}$$

These values are to be compared with those obtained by the Rayleigh-Ritz type solution, namely, $V/(b\omega) = 2.80$; $\omega_f = 67.4$ radians per sec; $V_f = 385$ mph.

It appears then that although flutter actually occurs at values of reduced speed and frequency which differ slightly from those deduced by the approximate solution, the flutter speed computed by the Rayleigh-Ritz procedure is in excellent agreement with the true value.

Using Equations [34] and [35], together with Equations [61] to [64], the true shape of the wing at flutter can be studied. The values of the roots λ and the constants a, b, c, d for the flutter point, obtained in terms of a_{12} arbitrarily set equal to —1.0000, are given in Table 1. The corresponding values of the Y functions at several points along the wing span are given in Table 2.

Suppose, now, boundary conditions are imposed so that at the start of the motion the bending deflection y at the wing tip is unity, while the bending velocity is zero, i.e., at $t = 0$ and $l = 20$ ft; $y = 1.00$ and $\dot{y} = 0$. It is then found that the wing shape during this motion can be described with excellent accuracy by the expressions

$$y = \overline{Y}_1(x) \cos \omega t \qquad [34a]$$

$$\theta = \overline{Y}_3(x) \cos (\omega t - 56° 57') \qquad [35a]$$

where $\overline{Y}_1$ and $\overline{Y}_3$ are functions of the spanwise co-ordinate x, given graphically in Fig. 4. The significance of being able to replace Equations [34] and [35] by the two simpler relations given will be discussed in the "Discussion and Results" section of the paper.

TABLE 2 VALUES OF Y FUNCTIONS AT SEVERAL POINTS ALONG SPAN[a]

x	Y_1	Y_2	Y_3	Y_4
0	0	0	0	0
4	0.0933	—0.1561	0.1313	0.003298
8	0.3450	—0.5629	0.2532	0.002584
12	0.7089	—1.1340	0.3545	—0.002053
16	1.1352	—1.7902	0.4232	—0.006471
20	1.5827	—2.4738	0.4481	—0.003604

[a] Based on data given in Table 1.

It is of interest to compare the true wing shape at flutter with the shape implied by the approximate solution of the third section of the paper. The wing shape predicted by the approximate solution is described by Equations [5] and [6] together with Equations [23] to [26]. The constants entering into the last two of these equations are fixed in relation to each other by the homogeneous Equations [27] and [28]. If the motion is started as before, i.e., with unit bending deflection and zero bending velocity at the wing tip, the wing shape for the approximate solution is described by

$$y = \overline{Y}_b(x) \cos \omega t \qquad [5a]$$

$$\theta = \overline{Y}_t(x) \cos (\omega t - 56° 51') \qquad [6a]$$

where the functions $\overline{Y}_b$ and $\overline{Y}_t$ are given graphically in Fig. 4.

Comparison of Equations [34a] and [35a] with Equations [5a] and [6a] shows that the approximate solution predicts the phase relationship between the bending and torsional components of the motion almost exactly. Also, the spanwise distribution of bending, as obtained by the approximate and exact methods, is in almost perfect agreement. The Rayleigh-Ritz type solution, however, introduces a slightly larger component of torsional displacement than does the solution by exact methods.

The excellent accuracy with which the approximate solution predicts the wing shape at flutter accounts for the close agreement which exists between the true flutter speed and that calculated by the approximate method.

Further analysis of the findings of this section is deferred until later, under the heading "Discussion and Results."

VIBRATION MODES IN VACUO

Aside from the conditions at flutter, several interesting problems arise in connection with ground vibration testing, i.e., vibration of the structure in still air. Tests of this kind are performed to fix the frequencies and shapes of the predominant "ground modes." This information has various uses. Thus the ground modes can be used in Rayleigh-Ritz type flutter analyses, replacing the uncoupled modes of the third section of the paper (1, 12, 13). The ground modes can also be used to check the accuracy of uncoupled modes which have been derived by analytical methods. The latter modes are usually calculated during the design period of an aircraft, and an experimental check is always desirable. It will be shown in this section that the experimental ground modes and the calculated uncoupled modes

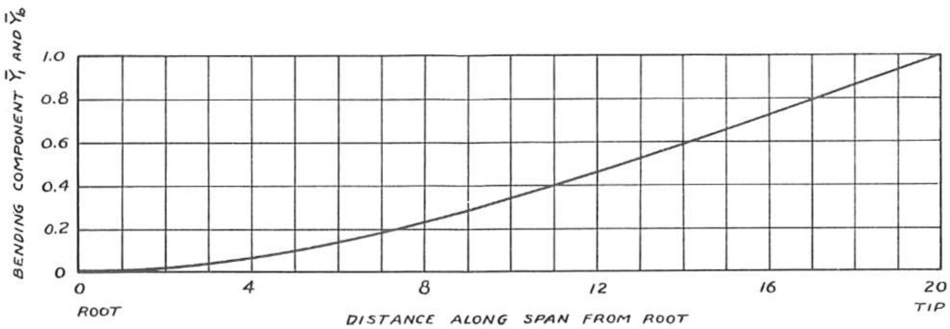


FIG. 4(a) SHAPE OF UNIFORM WING AT FLUTTER. SEE PAGE A-203: THE BENDING COMPONENT $\bar{Y}_1$, EQUATION [34a]; AND THE BENDING COMPONENT $\bar{Y}_b$, EQUATION [5a]

(The shape of $\bar{Y}_b$ is the same as that of the first uncoupled bending mode; differences in shape between $\bar{Y}_b$ and $\bar{Y}_1$ are negligibly small.)

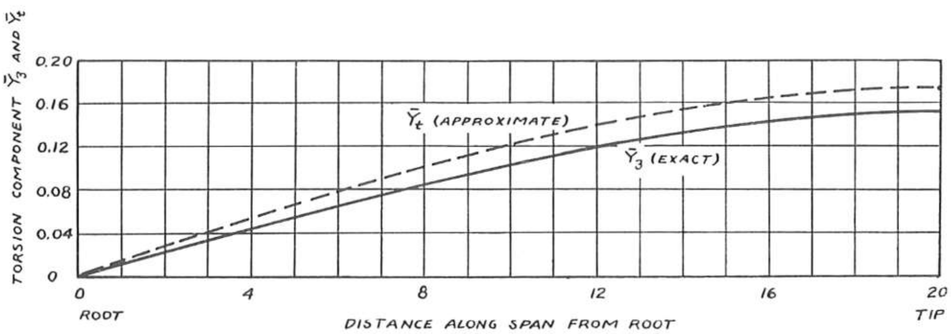


FIG. 4(b) SHAPE OF UNIFORM WING AT FLUTTER. SEE PAGE A-203: THE TORSION COMPONENT $\bar{Y}_3$, EQUATION [35a]; AND THE TORSION COMPONENT $\bar{Y}_t$, EQUATION [6a]

($\bar{Y}_3$ describes the true shape of the wing at flutter; $\bar{Y}_t$ is the corresponding torsional displacement predicted by the approximate solution of the Rayleigh-Ritz type.)

have several similar properties which can be used as cross checks.[6]

It becomes of interest to relate the characteristics of the ground and uncoupled modes, to see what properties they have in common. Since the effects of the air forces acting at zero reduced speed are small, it is permissible to treat both the uncoupled and ground modes as vibrations of the system in vacuo. It is recognized then that the ground modes closely resemble the normal modes for the wing structure. The properties of normal modes are well known, and in fact the findings of this section are largely restatements of several important theorems of classical dynamics

The exact solution for the flutter of a uniform wing reduces to the problem of vibration in vacuo if the air-force coefficients L_y, L'_y, ..., M_θ, M'_θ are set equal to zero. By reference to Equations [44] to [47], with the air-force terms deleted, it can be shown that

$$\frac{Y_1}{Y_2} = \frac{Y_3}{Y_4} \qquad [69]$$

Equations [34] and [35] are then reducible to the forms

$$y = Y_1(x) \cos(\omega t + \psi') \qquad [70]$$

$$\theta = Y_3(x) \cos(\omega t + \psi') \qquad [71]$$

where the complete expressions for Y_1 and Y_3 are given by

$$Y_1 = \sum_r \{A_r e^{\lambda_r x} + A_{r+1} e^{-\lambda_r x}\} \qquad [72]$$

$$Y_3 = \sum_r \{C_r e^{\lambda_r x} + C_{r+1} e^{-\lambda_r x}\} \qquad [73]$$

where

$$r = 1, 3, 5$$

It is to be noted that Equations [70] and [71] do describe a normal oscillation, with all the wing components either in or directly out of phase. The solution allows three values of λ_r; one of these is found to be real, while the remaining two are imaginary.

An outline of the exact solution to determine the normal modes of the uniform wing follows the ideas of the fourth section closely. Thus Equations [70] and [71] are substituted in the equations of motion. Y_1 and Y_3 are then taken in the exponential forms of Equations [40] and [42], resulting in a set of two homogeneous equations in the constants A and C. (These are the same as Equations [44] and [46], with the air-force terms omitted.) The determinantal equation arising from this homogeneous set fixes three values of λ_r for each chosen value of ω. The complete forms for the Y functions are then given by Equations [72] and [73]. The boundary conditions for the Y_1 and Y_3 functions are the same as those given by Equations [54] and [55]. These six homogeneous equations can be written in terms of only six constants of the type A_r; for a valid solution it is necessary that the determinant Δ' formed from the six equations be zero. Clearly, the determinant Δ' and the determinant Δ of Equation [68] are related to each other.

The natural frequencies and mode shapes of the first two normal modes for the wing described in the fifth section have been determined. The schedule of computations followed the methods of that section: An arbitrary value of ω was chosen; the corresponding values of λ_r and Δ' were computed; Δ' was then plotted versus ω, as shown in Fig. 5. Each time the curve

[6] Methods are also available to "uncouple" ground modes, yielding modes which involve only bending or torsional displacements. These are useful in flutter analyses in the manner outlined in the third section. The "uncoupling" procedure is the inverse of the one used later in this section to combine uncoupled modes.

crosses the $\Delta' = 0$ line, a normal mode is indicated. The mode frequency being fixed in this manner, Equations [70] to [73] can be used to describe the mode shapes.

It is seen that the first two normal modes for the uniform wing have natural frequencies of 45 radians per sec and 89.7 radians per sec. Plots of the functions $Y_1(x)$ and $Y_3(x)$ for the two modes are given in Figs. 6(*a*) and 6(*b*). The fundamental mode is predominantly bending, while the higher mode has a large torsion component. For this reason the bending component of the first mode is shown normalized at the wing tip; for the second mode, the torsion component is normalized.

The normal mode frequencies 45 radians per sec and 89.7 radians per sec are to be compared with the uncoupled frequency of 50 radians per sec for fundamental bending and 87 radians per sec for fundamental torsion. As noted in the captions of Figs. 6(*a*) and 6(*b*), the shapes of the normal and uncoupled modes bear striking resemblances to each other. Thus the bending component of the first normal mode has a spanwise distribution which is almost identical with that of the first uncoupled bending mode; a similar agreement in shape exists between the fundamental uncoupled torsion mode and the torsion component of the second normal mode. It is to be noted, however, that the remaining components of the two normal modes have shapes which differ greatly from those of the two uncoupled modes.

Suppose, now, that the two uncoupled bending and torsion modes are "coupled." The "coupled" frequencies which result from this combination can be found by solving Equation [29], with the air-force terms deleted. The coupled modes found in this way have natural frequencies of 49 radians per sec and 92.8 radians per sec. The proximity of these values to the true normal mode frequencies is to be noted. Furthermore, it is clear from the previous discussion that differences in shape between the coupled modes and the true modes for the wing will be minor.

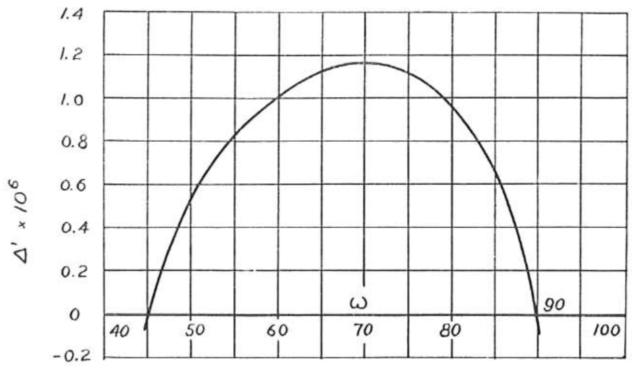


FIG. 5 DETERMINANT Δ' PLOTTED VERSUS ω FOR GROUND MODE SOLUTION

(Each time Δ' equals zero, a normal mode is indicated.)

It is to be expected, then, for the uniform wing, that the true ground modes and the uncoupled modes can be used interchangeably in flutter analyses of the Rayleigh-Ritz type. Also, comparison of calculated uncoupled modes with experimentally determined ground modes should show frequency and mode-shape checks. If a ground and an uncoupled mode have comparable natural frequencies, their corresponding components of displacement should have spanwise distributions which are in good agreement with each other.

FORCED OSCILLATIONS IN FLIGHT

Associated with the problem of determining the nature of the free wing oscillations at flutter is an analysis of the response of a wing when forced to oscillate while in flight. One experimental method for finding the flutter speed of a craft is to fly the airplane at gradually increasing speeds, at the same time noting the wing responses due to forced excitation over a range of frequencies

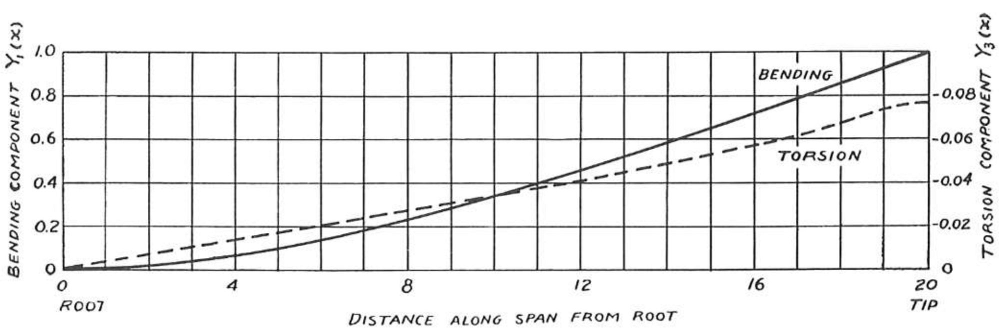


FIG. 6(*a*) SHAPE OF FIRST NORMAL MODE FOR UNIFORM CANTILEVER WING; FOR VIBRATION IN VACUO; NATURAL FREQUENCY $\omega = 45$ RADIANS PER SEC

(The shape of the fundamental uncoupled bending mode for the wing is the same as the bending component in this mode.)

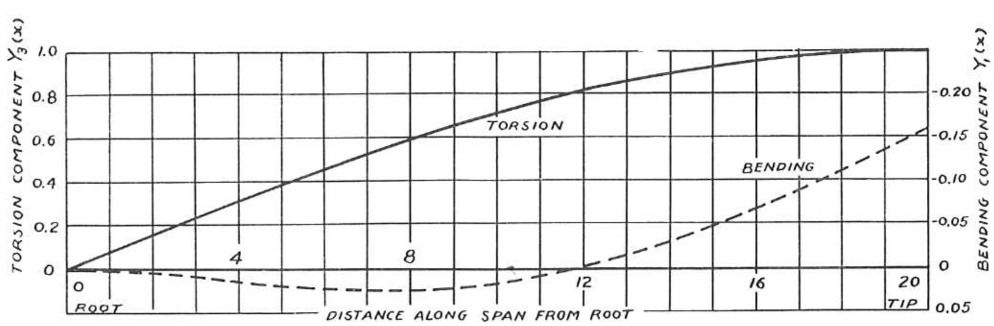


FIG. 6(*b*) SHAPE OF SECOND NORMAL MODE FOR UNIFORM CANTILEVER WING; FOR VIBRATION IN VACUO; NATURAL FREQUENCY $\omega = 89.7$ RADIANS PER SEC

(The shape of the fundamental uncoupled torsion mode for the wing is the same as the torsion component in this mode, save for minor departures.)

(14, 15). When the responses grow toward an infinite resonance point, the approach of the critical speed is indicated. In connection with this method it is of interest to compare the shape of a wing forced to oscillate with that of the flutter mode.

Consider the forced vibration of a uniform cantilever wing near flutter. An outline of the solution to this problem will now be given, together with the results of a typical calculation for the wing described in the fourth section. To shorten the work, a simple, type excitation is assumed; a vertical force of the form $F \cos \omega t$ is applied to the wing tip at the elastic axis. The extension of the methods to cover other types of excitations will become apparent from the discussion which follows.

Clearly, for the forced-oscillation problem, the airplane speed V and the frequency of excitation ω (and consequently the reduced speed $V/b\omega$) are parameters which can be fixed at will; for practical reasons it is only necessary that the speed V be kept below the flutter speed. With this restriction, the free motion of the wing will always consist of convergent oscillations and attention need be given only to the response induced by the applied force.

It is readily seen that Figs. 2(a) and 2(b) continue to be applicable to describe the forces and moments acting on a small spanwise element of length of the vibrating wing. The equations of motion for the wing then remain identical in form to Equations [32] and [33], so long as the changed significance of ω and V is kept in mind. The wing shape can accordingly be described by forms similar to Equations [34] and [35], together with Equations [40] to [43], in which case the satisfaction of the homogeneous Equations [44] to [47] remains a necessity.

The boundary conditions for the Y functions, however, require some alteration. In particular, for the excitation assumed, the transverse shearing force at the wing tip must be set equal to the vertical force applied at that station. The wing-tip bending moment must remain zero at all times; since the vertical force is applied at the elastic axis, the torsional moment at the wing tip must likewise be zero at all times. Setting the phase angle ψ in Equations [34] and [35] equal to zero, the boundary conditions for the Y functions become

$$\text{at } x = 0, \qquad Y_1 = Y_1^{\text{I}} = Y_2 = Y_2^{\text{I}} = Y_3 = Y_4 = 0 \ldots\ldots [74]$$

$$\text{at } x = l, \qquad Y_1^{\text{II}} = Y_2^{\text{II}} = Y_2^{\text{III}} = Y_3^{\text{I}} = Y_4^{\text{I}} = 0 \ldots\ldots\ldots [75]$$

$$Y_1^{\text{III}} = -F/(EI_b)$$

The twelve simultaneous equations arising from conditions, Equations [74] and [75], together with the homogeneous Equations [44] to [47], completely determine the response of the uniform wing to the excitation applied at the wing tip.

To calculate the wing shape, the twelve values of λ_r and the corresponding ratios B_r/A_r, C_r/A_r, and D_r/A_r, are calculated from Equations [44] to [47]. The Y functions are then written in the forms Equations [61] to [64], and these are substituted in the boundary conditions, Equations [74] and [75]. Using transformations of the type Equation [67], these twelve simultaneous equations are solved for the constants a, b, c, and d. Equations [61] to [64] then describe the response of the wing to the applied excitation. It is to be noted that this response calculation is direct; it does not involve a trial-and-error procedure as did the finding of the flutter point.

The response of the uniform wing described in the fourth section to a vertical force of the form $F \cos \omega t$, applied to the wing tip at the elastic axis, has been calculated for the point

$$\frac{V}{b\omega} = 2.94; \quad \omega = 64.5 \text{ radians per sec}$$

The airplane speed corresponding to these values of reduced speed and frequency is

$$V = 388 \text{ mph}$$

This speed is slightly below the flutter speed for the wing.

The values of the four Y functions at several points along the span, calculated for a value of F equal to $-EI_b$, are given in Table 3. A study of these data shows that the torsional deflection of the wing has a shape similar to that observed for the flutter mode, but the bending deflection does not. In particular, the torsional deflection can be written as

$$\theta = \overline{Y}_3'(x) \cos (\omega t - 84°10') \ldots\ldots\ldots\ldots\ldots\ldots [76]$$

but the bending deflection cannot be described other than in terms of two Y functions, displaced 90 deg in phase relative to each other. The shape of the function $\overline{Y}_3'$ is found to be the same as that of $\overline{Y}_3$, Fig. 4(b).

For the flutter mode, the torsional motion was found to lag the bending deflection by a phase angle of 56°57′. In order to facilitate a comparison between the wing shape at flutter and the shape of the forced response calculated here, the forced response in bending is written in the form

$$y = \overline{Y}_1'(x) \cos (\omega t - 27°13') + \overline{Y}_2'(x) \sin (\omega t - 27°13') \ldots [77]$$

TABLE 3 VALUES OF Y FUNCTIONS AT SEVERAL POINTS ALONG SPAN FOR FORCED-VIBRATION CALCULATION

x	Y_1	Y_2	Y_3	Y_4
0	0	0	0	0
4	94.271	68.721	4.158	73.347
8	329.520	234.226	10.058	140.996
12	634.505	434.332	17.138	196.254
16	942.413	606.611	23.186	232.580
20	1195.445	694.646	25.005	245.006

NOTE: The magnitude of the applied force is arbitrarily set equal to $-EI_b$.

The component $\overline{Y}_1'$ of the bending response thus leads the torsional motion by 56°57′; the second component $\overline{Y}_2'$ lags $\overline{Y}_1'$ by an angle of 33°03′.

The functions $\overline{Y}_1'$, $\overline{Y}_2'$, and $\overline{Y}_3'$ are shown graphically in Fig. 7. The magnitude of the wing response has been arbitrarily reduced so that the forced torsional response is of the same magnitude as the torsional motion at flutter shown in Fig. 4(b). If the forced bending response agreed with the bending motion at flutter, Fig. 4(a) and Fig. 7(a) should be identical. Although the two figures do not exactly correspond, it is seen that differences between them are not extreme.

It appears that in general the shape of the forced response agrees well with the motion at flutter. Minor differences are present in the bending shapes which are largely restricted to the region near the applied force.

In conclusion, it is to be noted that the applied force is out of phase with the bending component $\overline{Y}_1'$ by a lagging phase angle of 152°47′.

Discussion and Results

The most interesting aspect of the present paper is the presentation of an exact method for finding the flutter speed of a cantilever wing, of rectangular plan form, with uniform mass and stiffness characteristics along its span. With the exact solution known, it is possible to determine the extent of the error in the calculated flutter speed introduced by the use of the conventional methods for flutter analysis. In addition, considerable insight to the flutter phenomenon is gained by following the details of the exact mathematical development.

From the aerodynamic point of view, the solution presented in the third section of the paper is in need of further refinement. Thus in writing the air forces, Equations [15] and [16], it has

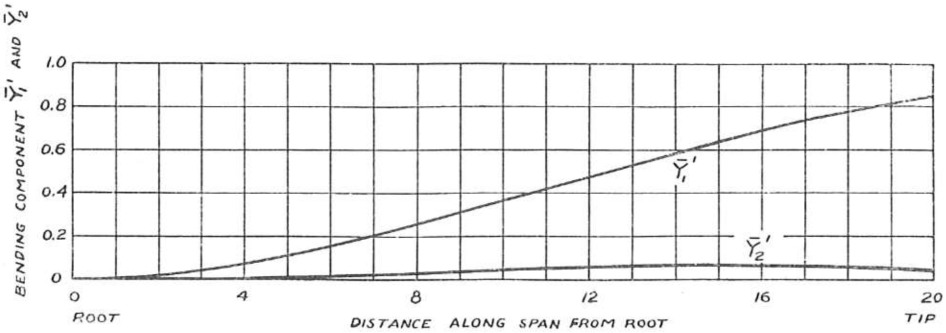


FIG. 7(a) SHAPE OF UNIFORM WING UNDER FORCED EXCITATION NEAR FLUTTER; THE BENDING COMPONENTS $\overline{Y}_1'$ AND $\overline{Y}_2'$; SEE EQUATION [77]

(The excitation consists of a sinusoidally varying vertical force applied to the wing tip at the elastic axis.)

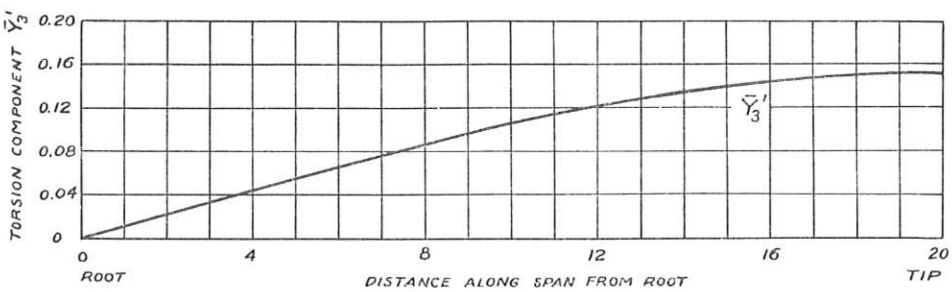


FIG. 7(b) SHAPE OF UNIFORM WING UNDER FORCED EXCITATION NEAR FLUTTER; THE TORSIONAL COMPONENT $\overline{Y}_3'$; SEE EQUATION [76]

(The shape of $\overline{Y}_3'$ does not differ appreciably from that of $\overline{Y}_3$, Fig. 4(b). The intensity of the forced excitation has been adjusted so that $\overline{Y}_3'$ and $\overline{Y}_3$ have equal magnitudes.)

been assumed that the fluid is incompressible and that two-dimensional flow conditions exist at each wing station. Actually, air is a compressible fluid and the aerodynamic wing loading is somewhat modified by the effects of induced downwash. These considerations have been omitted from the present analysis, since their introduction causes the solution to the dynamic problem to become of extreme complexity and hardly tractable. Investigation of these effects by others have shown that they influence the flutter speed of the wing described in the fourth section to only a minor extent.

Certain details of the exact solution presented in the third section are of particular interest. For example, the general nature of the forms taken to describe the wing shape at flutter, Equations [34] and [35], should be noted. The wing bending is written in terms of two Y functions, which are dependent only upon the spanwise co-ordinate x, and which are 90 deg out of phase with respect to each other. Since the two Y functions are not necessarily of the same form, allowance is made for the existence of a "traveling wave" in the wing-bending shape at flutter. Thus all stations must not necessarily pass through a zero bending deflection at the same instant, as is assumed in the conventional methods of flutter analysis. Similar remarks apply to Equation [35] which describes the torsional displacement of the wing.

The results of the calculations of the fourth section indicate that this general treatment of the wing shape is unnecessary. It is shown that, to a very excellent approximation, the wing shape can be described by a bending mode, with each station moving sinusoidally in phase with every other station, together with a similar torsion mode. The bending and torsion modes are displaced relative to each other by a phase angle whose magnitude is dependent on the characteristics of the wing being studied.

A further enlightening point is brought out by the calculations of the fourth section. It appears that the bending shape of the wing at flutter is practically identical with the shape of the fundamental uncoupled bending mode of the wing when vibrating in vacuo, Fig. 4(a). Also, the torsion mode at flutter, Fig. 4(b), does not differ appreciably in shape from that of the fundamental uncoupled wing torsion mode.

In view of these findings, it is not surprising to discover that the results of a Rayleigh-Ritz type flutter solution, based on the fundamental bending and torsion uncoupled modes for the wing, agree well with the results of the exact solution. In particular, comparing the flutter speeds and wing shapes as calculated by the methods of the third and fourth sections, it is found that:

1 The flutter speed calculated by the approximate method is in close agreement with the true value.

2 The bending shape of the wing at flutter is the same when determined by the two methods. For equal bending deflections the Rayleigh-Ritz type solution introduces slightly greater torsional deformation than does the exact solution.

3 The approximate solution predicts the phase angle between the bending and torsional motions at flutter to a high degree of accuracy.

It is apparent from these results that a Rayleigh-Ritz type solution based on the fundamental uncoupled wing modes is entirely adequate for the flutter analysis of a uniform cantilever wing. Since Rayleigh-Ritz procedures apply with equal facility to wings of nonuniform character, it is probable that the accuracy of such approximate methods does not suffer appreciably as the restriction of uniformity of the wing structure is relaxed.

The relation between the uncoupled modes and the normal modes for the uniform wing is discussed in the sixth section. It has already been mentioned that the characteristics of the normal modes, which are excited during free vibrations of the wing in vacuo, do not differ materially from those of the predominant modes observed during ground vibration tests. It is further shown in the same section that for the uniform wing studied, the bending component of the first normal mode has a shape which is identical with that of the fundamental uncoupled bending mode, Fig. 6(*a*), while a similar agreement exists between the shape of the fundamental uncoupled torsion mode and the torsion component of the second normal mode, Fig. 6(*b*). The remaining bending and torsion components in the first two normal modes are of unimportant magnitude.

On this basis it may be concluded that the ground modes and the uncoupled modes can be used interchangeably in flutter analyses of the Rayleigh-Ritz type. Furthermore, the remarks of the preceding paragraph show that it should be possible to check the shapes of calculated uncoupled modes by comparing them with the proper components of experimentally determined ground modes. It is also shown in the sixth section that the natural frequencies for the two sets of modes should be in approximate agreement.

The method for determining the wing response due to forced excitation is outlined in the seventh section. The results of a typical response calculation are given, for the case where the excitation consists of a vertical force, of sinusoidally varying magnitude, applied to the wing tip at the elastic axis. The airplane speed and the frequency of excitation are taken at values slightly below those at flutter. It is found that for this case of forced excitation near flutter the shape of the induced response agrees fairly well with the wing shape at flutter. The differences in shape which do appear are restricted largely to the spanwise region near the point of application of the vertical force.

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Appendix

Solution of the Rayleigh-Ritz Equations of Motion [19] and [20] Without Introduction of Complex Forms

The solutions for y_0 and θ_0 are taken in real form as follows

$$y_0 = A \cos \omega t + B \sin \omega t \qquad [78]$$

$$\theta_0 = C \cos \omega t + D \sin \omega t \qquad [79]$$

where A, B, C, and D are constants. These expressions are substituted in the equations of motion, Equations [19] and [20], the terms in each equation with $\sin \omega t$ and $\cos \omega t$ affixed are separated, and the sets of terms are individually equated to zero. After some manipulation the following equations are obtained[7]

$$L_{11} A - L'_y B - 0.673\, L_{12} C - 0.673 L'_\theta D = 0 \qquad [80]$$

$$L'_y A + L_{11} B + 0.673\, L'_\theta C - 0.673 L_{12} D = 0 \qquad [81]$$

$$-M_{21} A - M'_y B + 0.772\, M_{22} C - 0.772 M'_\theta D = 0 \qquad [82]$$

$$M'_y A - M_{21} B + 0.772\, M'_\theta C + 0.772 M_{22} D = 0 \qquad [83]$$

It is seen that the nontrivial solutions, Equations [78] and [79], are possible only if the determinant formed by the coefficients of A, B, C, D in Equations [80] to [83] vanishes. Expanding this determinant by minors, it can be shown that it is identical with the expression

$$(\Delta_1 - \Delta_2)^2 + (\Delta_3 + \Delta_4)^2 \qquad [84]$$

The form [84] can now be factored and equated to zero, giving as the stability condition

$$[(\Delta_1 - \Delta_2) + i(\Delta_3 + \Delta_4)][(\Delta_1 - \Delta_2) - i(\Delta_3 + \Delta_4)] = 0 \qquad [85]$$

The vanishing of the left-hand bracket is clearly identical with the stability condition [29]. Since, at flutter, $(\Delta_1 - \Delta_2)$ and $(\Delta_3 + \Delta_4)$ must vanish simultaneously, it is evident that no additional flutter points are introduced by the presence of the right-hand bracket in Equation [85]. It is seen from this that the present method of solution is equivalent in every respect to the solution outlined in the third section of the paper.

It is important to note that the determinant [84] is an unsymmetric one, showing that the flutter problem and those of classical dynamics, for the free vibration of elastic systems in vacuo, are not similar. A brief discussion of systems governed by an unsymmetric stability determinant is given by Routh (7).

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[7] Notation for this discussion is carried over from the third section.